

Assignment 3: Trends and Cloud Services Overview

1. Benefits of Serverless Architecture

Serverless architecture allows developers to build and run applications without managing infrastructure. Benefits include:

- **Scalability:** Automatically scales with demand.
- **Cost-Efficiency:** Pay only for actual usage, reducing costs.
- **Reduced Operational Overhead:** No need to manage servers, allowing developers to focus on code.
- **Faster Time-to-Market:** Accelerates development and deployment cycles.

2. Progressive Web Apps (PWAs)

PWAs are web applications that offer native app-like experiences. Key features:

- **Offline Access:** Functionality without an internet connection.
- **Installability:** Can be added to the user's home screen.
- **Responsive Design:** Adapts to various screen sizes and devices.
- **Push Notifications:** Engage users with timely updates.

PWAs enhance user engagement and are cost-effective alternatives to native apps.

3. Role of AI and Machine Learning in Software Architecture

AI and ML are transforming software architecture by:

- **Enhancing User Experience:** Personalized recommendations and intelligent interfaces.
- **Automating Processes:** Predictive analytics and decision-making support.
- **Improving Security:** Anomaly detection and threat prediction.
- **Optimizing Performance:** Resource allocation and system tuning based on usage patterns.

Cloud Computing Service Models

- **Infrastructure as a Service (IaaS):** Provides virtualized computing resources over the internet.
 - *Use Case:* Flexible and scalable hosting solutions.
- **Platform as a Service (PaaS):** Offers a platform allowing customers to develop, run, and manage applications.
 - *Use Case:* Streamlined application development without managing underlying infrastructure
- **Software as a Service (SaaS):** Delivers software applications over the internet on a subscription basis.

- *Use Case:* Accessible and maintenance-free software solutions for end-users.